

# Aditya Barman

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## EDUCATION

### University of Illinois at Urbana-Champaign (UIUC)

Champaign, IL

*Bachelor of Science in Computer Science & Statistics, Minor in Mathematics*

*Expected Graduation: May 2027*

- GPA: 4.0/4.0
- James Scholars Honors Program
- Activities: Pulse Competitions Committee, CS Sail Logistics Staff, Simplify Campus Ambassador

## EXPERIENCE

### Motorola Solutions

Chicago, IL

*Software Engineering Intern & Co-Op*

*May 2025 – Present*

- Designed and deployed a full-stack RAG system using LangChain, ChromaDB, HuggingFace embeddings, and Gemini API, indexing 30k+ proprietary documents with vector storage and API integration to deliver structured knowledge retrieval inside Motorola's documentation platform.
- Implemented ingestion pipelines with chunking, embeddings, and access controls, improving initial retrieval precision by over 40% and boosting accuracy by 65% through JSON serialization of diagrams.
- Built a Google Drive API crawler to populate the vector store, reducing indexing latency from hours to seconds and scaling to 250+ engineers' workspaces.
- Extended RAG system with multimodal capabilities using OpenAI CLIP embeddings to answer schematic questions and automated rack/cable diagram generation with Python and DrawSVG to cut manual effort by 80%.

### University of Illinois & United States Department of Agriculture (USDA)

Urbana, IL

*Computer Vision & Machine Learning Researcher*

*Nov. 2024 – May 2025*

- Engineered an OpenCV-based image processing pipeline to warp, segment, and analyze 198 kernel/flake image pairs, extracting area and shape features with a 95%+ success rate.
- Implemented feature extraction algorithms to quantify kernel to flake surface areas ratios, producing a dataset of 1,700+ measurements for downstream ML.
- Applied TensorFlow regression models to predict popcorn flake expansion from kernel characteristics, improving predictive accuracy and accelerating breeding program analysis.

### Chief Digital and Artificial Intelligence Office – Department of Defense

Washington, D.C.

*Software Development Intern – AI4Defense Program*

*June 2024 – Aug. 2024*

- Developed a Python application with OpenAI API integration to extract and analyze key terms from government documents, automating workflows and reducing manual effort by 90%.
- Built a full-stack prototype supporting PDF ingestion, frequency analysis, sentiment scoring, and visualization for rapid policy assessment by government officials.
- Presented solution at the AI4Defense Showcase and achieved a podium finish.

### George Mason University (GMU)

Fairfax, VA

*Artificial Intelligence Research Intern*

*June 2024 – Aug. 2024*

- Compared 4 large language models' (ChatGPT, Copilot, Claude, Gemini) performance in grading algorithm assignments to assess the feasibility of LLM evaluation.
- Computed Intraclass Correlation Coefficients and other statistical metrics to quantify consistency across models.
- Co-authored paper accepted to the MIT Undergraduate Research Technology Conference (URTC) 2024.

## PROJECTS

### EchoBrief | *FastAPI, Python, JavaScript, Google Gemini API, OpenAI Whisper*

July 2025

- Developed an AI-powered Post-Incident Analysis Tool at Motorola Solutions' Open Innovation Hackathon, processing radio traffic recordings to generate structured after-action dashboards.
- Integrated OpenAI Whisper for transcription and Google Gemini API for summarization + event extraction, surfacing officer contributions, key incidents, and overall timeline with timestamps and floorplan overlays.
- Delivered incident summaries and visual event mapping that reduced manual after-action review time by 70%.

### LearnLion | *Python, Flask, React, HTML/CSS, Google Gemini API*

Nov. 2024

- Won second place at HackIllinois x CS124 Hackathon for a tutoring chatbot aimed at elementary students.
- Implemented a custom substring frequency mapping algorithm to generate accurate, context-aware responses.

## TECHNICAL SKILLS

**Languages:** Python, C++, Kotlin, Java, SQL (Postgres), JavaScript, HTML/CSS, R, MATLAB

**Frameworks:** Flask, FastAPI, LangChain, Tensorflow, Django, Express

**Developer Tools:** Git, Node.js, MongoDB, Docker, AWS (EC2, S3, Lambda, RDS, etc.), Google Cloud Platform (GCP)

**Libraries:** React, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, OpenCV, HuggingFace Transformers